Retrieval, Association, and Shrinkage in BundleFuse

What the connections to established methods actually explain

The connection to text retrieval begins with a simple observation: a historical bundle is a collection of items, just as a document is a collection of terms. An observed partial bundle can therefore be matched against historical bundles using occurrence, frequency and size information. Completion then requires an additional step: the matched bundles transfer their support to candidate items. The equations below make this interpretation exact for the binary, unit-weight training bundles used in the full-ranking POG formulation.

1 Define the observations and the retrieval analogy

Let $\mathcal{I} = \{1, \dots, n\}$ be the item catalogue, and let m training bundles be indexed by $e \in \{1, \dots, m\}$. Define the binary incidence matrix $B \in \{0,1\}^{m \times n}$ by

$$B_{ei} = \begin{cases} 1, & \text{item } i \text{ occurs in training bundle } e, \\ 0, & \text{otherwise.} \end{cases}$$

An item contributes at most once to a training row. Define its training frequency $f_i = \sum_{e=1}^{m} B_{ei}$ and bundle size $d_e = \sum_{i=1}^{n} B_{ei}$. Let $S \subseteq \mathcal{I}$ be a nonempty observed set, let $c \in \mathcal{I} \setminus S$ be a candidate, and let $x_S \in \{0,1\}^n$ be the indicator of S.

Text retrieval	Bundle completion
Document	Historical training bundle e
Term	Catalogue item i
Occurrence of a term in a document	Binary entry B_{ei}
Number of documents containing a term	Number of bundles containing an item, f_i
Document length	Bundle size d_e
Query terms	Observed items S

In a basic TF–IDF formulation, term i in document e receives weight

$$\mathrm{tfidf}(i,e) = \mathrm{tf}(i,e)\,\mathrm{idf}(i), \qquad \mathrm{idf}(i) = \log \frac{m}{f_i} \quad (f_i > 0). \tag{1}$$

Here tf counts occurrences within the document, while idf discounts terms appearing in many documents. Common variants smooth the logarithm and normalize document vectors by their Euclidean length [1]. With binary bundle rows, the corresponding occurrence quantity is simply B_{ei}; repeated occurrences within a row are not modeled.

This analogy motivates the roles of the weights. It does not determine their exact functional form. BundleFuse uses inverse square roots and explicit offsets, as defined next, rather than the logarithmic IDF above.

2 Match the observed items to historical bundles

Define an observed-item weight a_i and a historical-bundle weight v_e by

$$a_i = \frac{1}{\sqrt{f_i + 5}}, \qquad v_e = \frac{1}{\sqrt{d_e + 1}}. \tag{2}$$

The historical bundle's match score is

$$\boxed{o_e(S) = v_e \sum_{s \in S} B_{es} a_s = \frac{1}{\sqrt{d_e + 1}} \sum_{s \in S} \frac{B_{es}}{\sqrt{f_s + 5}}.} \tag{3}$$

Each summand corresponds to one observed item actually present in the historical bundle. Its frequency discount is larger for a rarer observed item. The outer factor reduces the contribution of the same match when it occurs inside a larger bundle. Multiple matches add together.

For example, an observed item appearing in four training bundles has weight $1/\sqrt{9} = 1/3$, whereas one appearing in 95 bundles has weight $1/\sqrt{100} = 1/10$. Within the same historical bundle, the former match contributes $10/3$ times as much. Likewise, the bundle-size factors for sizes three and fifteen are $1/2$ and $1/4$, respectively: the same weighted overlap receives twice as much weight in the smaller bundle. These are illustrative calculations, not observations about a particular evaluation query.

What is shared with TF–IDF? Both constructions discount a clue according to how widely it occurs in the training collection. Document-vector normalization can additionally reduce length effects; IDF alone does not normalize document length. This is a parallel in weighting principles, with explicit differences in the formulas:

- The item-frequency factor $(f_i + 5)^{-1/2}$ is a smoothed decreasing frequency weight, rather than logarithmic IDF.
- The factor $(d_e + 1)^{-1/2}$ depends on binary bundle size. It is not the Euclidean normalization of a TF–IDF-weighted document vector, whose norm would also depend on its individual term weights.
- Frequency weights act on the observed query items in (3). Candidate items do not receive that same divisor when votes are transferred.

The offsets 5 and 1 and the square-root exponents are fixed modeling choices. The analogy explains their roles, not the optimality of those numbers. There is a further distinction: if $f_i = m$, basic unsmoothed IDF is zero, whereas $a_i = (m + 5)^{-1/2}$ remains positive. Nor is a_i a logarithmic information content. Thus the direct score is best described as *frequency-discounted historical retrieval*, not as TF–IDF itself. These are discounts per recorded match; repeated matches can still accumulate large total support. They do not establish complete removal of popularity effects.

3 Transfer historical-bundle matches to completion candidates

A historical bundle proposes its member items as possible completions. Accordingly, sum its match score into candidate c only if it contains c:

$$\boxed{p_c(S) = \sum_{e=1}^{m} B_{ec} o_e(S) = \sum_{s \in S} a_s \sum_{e=1}^{m} v_e B_{ec} B_{es}.} \tag{4}$$

The inner sum is the size-discounted number of bundles containing both the candidate and observed item s. The outer sum combines this support across the observed items, with their individual frequency discounts.

Set $D = \mathrm{diag}(a_i) \in \mathbb{R}^{n\times n}$ and $V = \mathrm{diag}(v_e) \in \mathbb{R}^{m\times m}$. The complete calculation is

$$o(S) = VBDx_S, \qquad \boxed{p(S) = B^\top VBDx_S.} \tag{5}$$

Reading from right to left: weight the observed items, match them to historical bundles, discount bundle size, and transfer the bundle scores to their members. This provides a retrieval interpretation of the two incidence multiplications. The first multiplication produces historical-bundle evidence; the second converts that evidence into a completion ranking.

The formula retains self-contributions at observed-item entries of p. Candidates are restricted to $c \notin S$ when recommendations are returned. The displayed equation is the raw direct score, before catalogue standardization or combination with other sources.

4 Distinguish simultaneous matches from separate matches

Additive overlap gives one contribution for each observed item. To reward several observed items appearing in the same historical bundle, define

$$k_e(S) = \sum_{s\in S} B_{es} = |e \cap S|, \qquad [z]_+ = \max(z, 0), \qquad \beta = 1.5.$$

The incidence-enhanced score is

$$\boxed{g_c(S) = \sum_e B_{ec} o_e(S)\big[1 + \beta[k_e(S) - 1]_+\big].} \tag{6}$$

For one matching observed item the multiplier is one. For two it is 2.5; for three it is four. If there are no matches, $o_e(S) = 0$. This amplification applies separately to each historical bundle before its vote is transferred. The setting $\beta = 1.5$ specifies the chosen strength.

Why retain this operation? A candidate may occur with s in one historical bundle and with t in another, without ever occurring with both together. Equation (6) distinguishes that situation from a historical bundle containing s, t, c. To see the distinction algebraically, define

$$T_{cst} = \sum_e v_e B_{ec} B_{es} B_{et}.$$

This is the weighted count of bundles containing all three distinct items. For $c \notin S$, the binary-query expansion of (6) is

$$g_c(S) = p_c(S) + \beta \sum_{\{s,t\}\subseteq S} (a_s + a_t) T_{cst}, \tag{7}$$

where each unordered pair of distinct observed items appears once. Thus direct matching uses candidate–seed pairs, while the correction uses candidate–seed–seed triples. It rewards supported pairs within S; it does not require the entire observed set to have appeared together.

The TF–IDF analogy explains the frequency and size discounts. The simultaneous term extends the scoring rule beyond additive item matching. Its triple counts cannot in general be reconstructed from the item-pair count matrix alone. This is a statement about information retained by these particular operators, not a claim that multi-item retrieval or higher-order recommendation is new.

5 Interpret PPMI as conditional occurrence plus inverse-frequency information

The PPMI source uses another normalization of the same training observations. Define

$$C = \operatorname{off}(B^\top B), \qquad q_i = \sum_j C_{ij}, \qquad T = \sum_{i,j} C_{ij}, \tag{8}$$

where off replaces a square matrix's diagonal by zero. C_{ij} counts training bundles containing both distinct items i, j. Assume $T > 0$. Sampling an ordered within-bundle pair from all recorded pair occurrences gives

$$P(i,j) = \frac{C_{ij}}{T}, \qquad P(i) = \frac{q_i}{T}.$$

The endpoint marginal q_i is not the bundle frequency f_i:

$$q_i = \sum_e B_{ei}(d_e - 1). \tag{9}$$

An item in a larger bundle participates in more pairs. This distinction must be retained when connecting the log-ratio to inverse frequency.

For a recorded pair (c, s), its pointwise mutual information is

$$\operatorname{PMI}(c,s) = \log \frac{P(c,s)}{P(c)P(s)} = \log \frac{C_{cs}T}{q_c q_s} \tag{10}$$

$$= \underbrace{\log \frac{C_{cs}}{q_c}}_{\log P(s|c)} + \underbrace{\log \frac{T}{q_s}}_{-\log P(s)}. \tag{11}$$

The first term measures how often s appears as a partner of c; the second is the inverse-frequency information of s in the ordered-pair distribution. The sum is positive exactly when s is more likely beside c than under its overall endpoint frequency. This is the precise information-theoretic parallel to discounting common terms.

Define positive PMI by $M_{cs} = [\operatorname{PMI}(c,s)]_+$ for recorded pairs, and zero for absent pairs. The raw source score is

$$r_{\text{pmi}}(c \mid S) = \sum_{s \in S} M_{cs}. \tag{12}$$

The implementation adds small numerical stabilizers to the denominators and logarithm. Equation (11) describes the underlying ratio before those stabilizers. Clipping removes negative associations from this source; it does not convert them into penalties. Unlike TF–IDF, PPMI includes the conditional association of two items, rather than only a term's frequency weight.

6 From association to similarity of association profiles

The direct PPMI entry M_{cs} asks whether c and s occur together above the independent-endpoint baseline. A different question is whether they have similar partners. This distinction explains the SVD source.

Write $M = U\Sigma V^\top$ for a singular value decomposition, with singular values in decreasing order. Let r be the retained dimension, and V_r the first r right singular vectors. Define

$$Z = MV_r = U_r\Sigma_r, \qquad u_i = \frac{Z_i}{\|Z_i\|_2 + 10^{-8}}, \qquad r_{\text{svd}}(c \mid S) = u_c^\top \frac{1}{|S|} \sum_{s \in S} u_s.$$

Here Z_i is the row for item i, not a singular vector. Orthogonality of the columns of V_r does not make these rows unit length. Before row normalization,

$$ZZ^\top = MV_rV_r^\top M^\top = U_r\Sigma_r^2U_r^\top. \tag{13}$$

If all singular directions are retained, its (c, s) entry is $\sum_j M_{cj}M_{sj}$. Thus it rewards sharing positively associated partners, even when $M_{cs} = 0$. Truncation retains the dominant directions in these association profiles. It is not the rank-r reconstruction $U_r\Sigma_rV_r^\top$ used directly as an item–item score.

The connection is to the use of SVD in latent semantic indexing [2]: compare objects through compressed occurrence profiles. Here the input is item–item PPMI, not a term–document table. The exact rank-r SVD minimizes $\|M - A\|_F^2$ over matrices A of rank at most r, where $\|\cdot\|_F^2$ sums squared entries. The numerical procedure approximates that decomposition. Neither this approximation criterion nor Equation (13) guarantees better recommendation rankings.

This also limits the interpretation: similar partners may indicate that two items are substitutes. Their similarity is not proof that they belong together. The source supplements observed co-membership rather than replacing its meaning.

7 From descriptions to deviations from a category mean

Association profiles require interaction observations. Supplied item descriptions provide another way to compare items. For one description modality, let $X \in \mathbb{R}^{n\times p}$ contain its row-normalized supplied vectors, with p feature coordinates. Let $\ell(i)$ be the supplied category label, and let $H \in \{0, 1\}^{n\times k}$ contain one indicator column for each of the k nonempty categories: $H_{ia} = \mathbf{1}[\ell(i) = a]$.

Define

$$\Pi = H(H^\top H)^{-1}H^\top, \qquad R = (I_n - \Pi)X. \tag{14}$$

I_n is the $n \times n$ identity matrix. Since $H^\top H$ contains the category sizes on its diagonal, $(\Pi X)_i$ is precisely the mean vector of all catalogue items in category $\ell(i)$. Moreover,

$$\widehat{A} = \arg\min_{A\in\mathbb{R}^{k\times p}} \|X - HA\|_F^2 = (H^\top H)^{-1}H^\top X, \qquad H^\top R = 0.$$

Thus category subtraction is exactly the residual of a least-squares fit using category indicators. It removes the part represented by a category-specific constant vector. This is a more precise connection than calling it "disentanglement."

Normalize each residual row, $E_i = R_i/(\|R_i\|_2 + 10^{-8})$. The descriptor score is

$$r_E(c \mid S) = E_c^\top \bar{E}_S, \qquad \bar{E}_S = \frac{1}{|S|}\sum_{s\in S} E_s.$$

It compares the candidate to the mean description of the observed items. This is the same centroid-matching operation that underlies the positive feedback part of Rocchio retrieval [3]; it is not the full Rocchio algorithm with an original query and negative feedback.

Row normalization generally destroys $H^\top E = 0$, even though $H^\top R = 0$. Consequently, only the unnormalized residual has the exact orthogonality property. Removing a category mean does not remove all category information or establish that remaining coordinates represent compatibility. For several modalities the operation is performed separately, followed by concatenation and normalization where required.

8 From descriptions of an item to descriptions of its partners

Two items can have related roles even if their own descriptions differ. The bridge incorporates the descriptions of historically connected items. Let E now denote the normalized concatenation of the available residual modalities. With D already defined as the seed-frequency diagonal, set

$$L = \operatorname{off}\Big(DB^\top \operatorname{diag}((d_e+1)^{-1})BD\Big), \qquad N = \mathcal{N}(LE), \qquad F = \mathcal{N}(E+\eta N), \quad \eta = 0.5,$$

where $\mathcal{N}$ divides each row by its Euclidean norm plus 10^{-8}. In particular, $(LE)_i = \sum_j L_{ij}E_j$ aggregates the residual descriptions of items sharing training bundles with i. L is symmetric but is not a row-stochastic transition matrix: its rows need not sum to one.

The bridge score is $r_F(c \mid S) = F_c^\top |S|^{-1} \sum_{s\in S} F_s$. Let A be the diagonal of the final row-normalization factors, so that $F = A(E+\eta N)$. Its similarity matrix expands exactly as

$$FF^\top = A(EE^\top + \eta EN^\top + \eta NE^\top + \eta^2 NN^\top)A. \tag{15}$$

The four terms compare own descriptions, own descriptions with partner profiles in either direction, and partner profiles with one another. This is a one-step graph aggregation followed by similarity matching. It provides a concrete route from descriptive resemblance to historical context; it is neither an iterated random walk nor a fitted graph neural network. The cross terms can be negative because category residual coordinates are signed.

9 From exact identities to pooled category or cluster relations

Rare items may have too few observations to estimate useful item-specific associations. Grouping items shares observations across their labels or feature memberships. This is pooling of counts, followed by lifting the resulting relation back to individual items.

For the category indicator H defined above, BH counts category occurrences per bundle. Write

$$A^{\mathrm{cat}} = BH, \quad C^{\mathrm{cat}} = \operatorname{off}((A^{\mathrm{cat}})^\top A^{\mathrm{cat}}), \quad h_a = \sum_e A^{\mathrm{cat}}_{ea}, \quad h_+ = \sum_a h_a.$$

For positive recorded counts define $Q^{\mathrm{cat}}_{ab} = [\log(C^{\mathrm{cat}}_{ab} h_+/(h_a h_b))]_+$, and use zero otherwise. Small numerical stabilizers are omitted here. The category association score is $(HQ^{\mathrm{cat}}H^\top x_S)_c$. The matrix products explicitly mean: collect seed labels, score relations between labels, then give each candidate the score of its label. These h_a are category-occurrence totals, not the endpoint totals of C^{cat}. This is a count-normalized log score; the exact independence interpretation established for item PPMI must not be transferred to it.

The POG category score also discourages categories that rarely repeat in one bundle. Let $n_a^{(1)}$ and $n_a^{(2)}$ count training bundles containing at least one and at least two items of category a, respectively. Set

$$\rho_a = \frac{n_a^{(2)} + 0.5}{n_a^{(1)} + 1}.$$

Subtract $0.5(1-\rho_a)$ for each seed of the candidate's category if $\rho_a < 0.15$. Under a Bernoulli model of repetition conditional on category presence, ρ_a is exactly the posterior mean with a $\mathrm{Beta}(1/2, 1/2)$ prior. The threshold and penalty magnitude are additional design choices, not consequences of that posterior mean.

Soft groups replace H by a matrix $J \in \mathbb{R}^{n\times k}$ of nonnegative feature-cluster memberships. Clusters are fitted by minibatch k-means to normalized concatenated supplied descriptors; membership is proportional to the exponential of similarity to a normalized cluster centre, with temperature

0.05. Each row sums approximately to one (the normalization includes a small stabilizer). For binary training bundles,

$$C^{\text{soft}} = \operatorname{off}(J^\top CJ), \qquad b_a = \sum_i J_{ia}, \quad b_+ = \sum_a b_a,$$

$$Q^{\text{soft}}_{ab} = \left[\log \frac{C^{\text{soft}}_{ab} b_+}{b_a b_b}\right]_+, \qquad r_{\text{soft}} = JQ^{\text{soft}} J^\top x_S,$$

using zero for absent counts and the same numerical qualifications. The identity $J^\top CJ = (BJ)^\top BJ - J^\top \operatorname{diag}(f)J$ explains the self-item subtraction: the same item must not supply both ends of a pair. Here b_a is catalogue membership mass, not a pair marginal.

This is relation sharing through groups. It is not a dot-product kernel in general: Q^{soft} need not be positive semidefinite. Given J, the soft count matrix contains no information beyond the item-pair matrix C. This contrasts with the incidence correction, which uses candidate–seed–seed counts from the original bundles. The selected full POG_DENSE recipe does not use the soft source; its role is to explain an available component, not an observed cause of the dense result.

10 From overlapping clues to diversity of support

The optional agreement score asks whether a candidate is supported by different observed items rather than nearly duplicate clues. For POG, define

$$W = \operatorname{off}(B^\top VBD), \qquad q_{cs} = W_{cs} + 0.15[E^t_c \cdot E^t_s]_+, \qquad A_c = \sum_{s \in S} q_{cs},$$

where E^t_i is the normalized category-residual title vector defined above. For distinct $s, t \in S$, set

$$K_{st} = \left[1 - 0.5\mathbf{1}[\ell(s) = \ell(t)] - 0.5[E^t_s \cdot E^t_t]_+\right]_+, \qquad K_{ss} = 0.$$

For $A_c > 0$, let $\pi_{cs} = q_{cs}/A_c$. These are nonnegative weights summing to one over seeds. Then

$$r_R(c \mid S) = \frac{\sum_{s,t} q_{cs} K_{st} q_{ct}}{A_c + 0.05} = \frac{A_c^2}{A_c + 0.05} \sum_{s,t} \pi_{cs}\pi_{ct} K_{st}. \tag{16}$$

When $A_c = 0$ the score is zero. The last sum is the expected dissimilarity of two independent seed draws from the candidate's support distribution. It has the quadratic-diversity form introduced by Rao [4]. It is multiplied here by a support-strength factor. A candidate supported by one seed has zero diversity; strong support spread across dissimilar seeds can receive more credit. "Independent draws" describes this mathematical sampling operation, not statistical independence of the original observations. No triangle inequality or entropy-concavity property of K is asserted.

This is not the incidence correction: it combines candidate–seed affinities and seed dissimilarities, without requiring a historical bundle containing all three items. It is also not a penalty for diversity of the recommended list. The selected full POG_DENSE recipe gives this source zero weight.

11 From heterogeneous scores to a regularized final estimate

The sources answer different questions and have different scales. For each raw source $r_h(i \mid S)$, define, over all n catalogue items,

$$\mu_h = \frac{1}{n}\sum_i r_h(i \mid S), \qquad \sigma_h^2 = \frac{1}{n-1}\sum_i (r_h(i \mid S) - \mu_h)^2, \qquad z_h(i \mid S) = \frac{r_h(i \mid S) - \mu_h}{\sigma_h + 10^{-9}}.$$

This is the full-ranking POG standardization, applied separately per query. It expresses scores relative to their catalogue mean and spread; it is not probability calibration. No held-out completion label enters these statistics. The full catalogue, including seeds before the final exclusion mask, is used.

Nonnegative scalar weights combine these standardized sources. They are chosen on validation data and need not sum to one. Therefore source fusion is not itself necessarily a convex average. For the selected full POG_DENSE recipe, write

$$Y_c = 0.5z_{\text{cat}} + 0.25z_{\text{bridge}} + 0.5z_g + 0.25z_{\text{pop}} + 0.25z_{\text{svd}}, \tag{17}$$

$$P_c = 0.5z_{\text{cat}} + 0.25z_{\text{bridge}} + 0.25z_{\text{pop}}, \qquad r_{\text{pop}}(c) = \log(1 + f_c), \tag{18}$$

where every z on the right is evaluated at $(c \mid S)$. The popularity source provides a query-independent frequency baseline before fusion, with logarithmic compression of large counts. Its increasing frequency weight has a different role from the decreasing weight on observed seed clues. P_c is a restricted fallback score, not an external probability distribution.

Let $s_0 \geq 0$ be its validation-selected scale and $\kappa > 0$ the frequency threshold parameter. The final score is

$$\boxed{\theta(c \mid S) = \frac{f_c Y_c + \kappa s_0 P_c}{f_c + \kappa}.} \tag{19}$$

This is exactly the unique solution of

$$\min_{t \in \mathbb{R}} \quad \frac{f_c}{2}(t - Y_c)^2 + \frac{\kappa}{2}(t - s_0 P_c)^2. \tag{20}$$

Differentiating gives $(f_c + \kappa)t = f_c Y_c + \kappa s_0 P_c$; the second derivative is strictly positive. This connects the gate to quadratic regularization [5], here toward a nonzero fallback. At zero frequency the fallback is used completely; at $f_c = \kappa$ the two scores receive equal weight; at large frequency Y_c dominates. The selected dense values are $\kappa = 0.5$ and $s_0 \simeq 1.46055$. With shrinkage disabled ($\kappa = 0$), the procedure returns Y_c directly, including for items of zero frequency.

The frequency-based coefficient imposes a reliance policy. It does not prove that the error variance of Y_c is proportional to $1/f_c$ or that the result is a calibrated Bayesian posterior. Also P_c and Y_c share sources. Writing $Y_c = P_c + H_c$ and $\rho_c = f_c/(f_c + \kappa)$ gives

$$\theta(c \mid S) = \big[s_0 + (1 - s_0)\rho_c\big]P_c + \rho_c H_c.$$

This is the precise separation between retained shared evidence and the frequency-controlled additional evidence. For the selected dense recipe, $H_c = 0.5z_g + 0.25z_{\text{svd}}$. Rare items are not automatically boosted: the direction of the change depends on the scores being mixed.

12 A coherent interpretation, with distinct mathematical roles

The resulting progression is:

1. Retrieve historical bundles using discounted overlap and transfer their support to candidates. TF–IDF supplies a related frequency-weighting principle.

2. Preserve simultaneous candidate–seed–seed occurrences where selected; these distinguish joint observations from a collection of separate pairs.

3. Share evidence when exact identity is too restrictive: compare association profiles, pool group relations, and compare descriptions and partner descriptions.

4. Standardize the resulting scores, select their contributions on validation, and regularize the combined estimate toward a restricted fallback using candidate frequency.

These connections explain what information each multiplication retains and how it affects the score. They do not derive every fixed constant from a common principle, establish a single joint optimization objective, or prove superiority of the whole estimator. In particular, rarity weighting, profile compression, mean removal, and quadratic regularization solve different mathematical problems. Their combination is the construction to assess, rather than a claim that all sources are versions of TF–IDF.

References

[1] Manning, Raghavan and Schütze (2008). *Introduction to Information Retrieval*, inverse document frequency. https://nlp.stanford.edu/IR-book/html/htmledition/inverse-document-frequency-1.html.

[2] Manning, Raghavan and Schütze (2008). *Introduction to Information Retrieval*, Chapter 18. https://nlp.stanford.edu/IR-book/pdf/18lsi.pdf.

[3] Manning, Raghavan and Schütze (2008). *Introduction to Information Retrieval*, Rocchio relevance feedback. https://nlp.stanford.edu/IR-book/html/htmledition/the-rocchio71-algorithm-1.html.

[4] C. R. Rao (1982). Diversity and dissimilarity coefficients: a unified approach. *Theoretical Population Biology* 21, 24–43. https://doi.org/10.1016/0040-5809(82)90004-1.

[5] Stanford STATS 305B. Penalized regression, ridge regression. https://web.stanford.edu/class/stats305b/notes/Ridge.html.